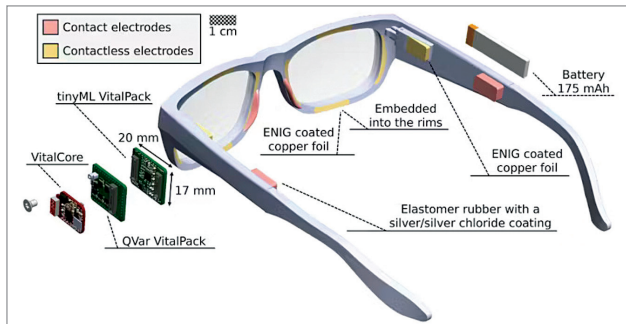


Smarter eye-tracking for smart glasses

ETH Zurich has introduced ElectraSight, an innovative eye-tracking system for smart glasses. Smart glasses are evolving rapidly, leveraging eye-tracking for seamless interaction with the world. Traditional systems use cameras,



These smart glasses have an efficient eye-tracking system (Credit: N. Scharer et al)

which are accurate but bulky, energy-intensive, and raise privacy concerns. It employs low-power sensors to detect tiny electrical signals generated by eye movements, ensuring high accuracy and efficiency. ElectraSight features three modules: data control, eye movement processing, and sensor management. Its compact AI model classifies six movements with 92% accuracy and ten with 81%, processing data in just 301 microseconds.

Hydrogen-ion batteries for sustainable energy storage

Scientists at UNSW Sydney have developed a hydrogen-ion battery, offering a sustainable and safer alternative to lithium-ion technology. An organic anode material, tetraamino-benzoquinone (TABQ), is central to its design, which supports efficient proton storage and rapid movement through hydrogen-bond networks, enhancing energy density and output. Unlike lithium-ion batteries, which pose fire risks and rely on resource-intensive materials, this innovation uses abundant light elements and operates effectively in sub-zero conditions. Demonstrating a lifespan of 3500 cycles, it holds promise for renewable energy storage and hydrogen transport, paving the way for scalable, eco-friendly energy solutions to combat climate change.

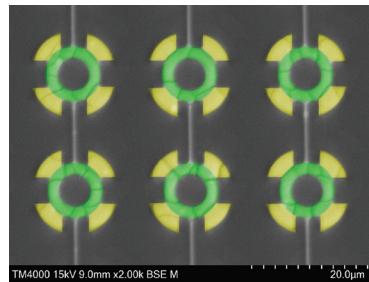
New alloy for better MRAM memory

Researchers at WPI-AIMR and Tohoku University have developed a cobalt-manganese-iron alloy thin film with exceptional perpendicular magnetic anisotropy (PMA) and high tunnel magnetoresistance (TMR), key traits for

MRAM. Gone are the days of two-megabyte floppy disks; modern data demands require advanced memory solutions. Magnetoresistive random access memory (MRAM), leveraging spintronics, offers low power consumption and high capacity. Unlike traditional iron-cobalt-boron alloys, this material excels in both properties, enabling stable, high-density data storage. Simulations confirm its potential for next-generation memory devices. This breakthrough advances spintronics-based storage technologies, paving the way for sustainable, high-performance memory solutions.

Self-assembly technique for 3D electronics

North Carolina State University researchers have developed a new self-assembly method for creating electronic components that shows great potential for advancing 3D chip



These microscopic objects were self-assembled using the new technique which the researchers refer to as a directed metal-ligand reaction or D-Met. So far, the technique can be used to create diodes, transistors and other simple electronic components (Credit: Julia Chang)

manufacturing. Unlike traditional processes, which rely on costly robotics and manual labour, this technique simplifies production by using a solution of carbon and oxygen ligands mixed with liquid metal particles. When poured into a mould, the ligands interact with ions, forming complex 3D structures like diodes and

transistors as the liquid evaporates. The process is faster, cheaper, and more consistent than conventional methods, reducing waste and defects while eliminating the need for expensive machinery. This breakthrough could revolutionise chip production by enhancing efficiency and yield.

Predicting material properties using limited data

Researchers at the Indian Institute of Science (IISc) and University College London (UCL) have advanced materials discovery using machine learning (ML) to predict material properties with minimal data. Leveraging transfer learning, they pre-trained models on large datasets and fine-tuned them for specific tasks, overcoming challenges posed by limited experimental data. The team optimised graph neural networks (GNNs) for 3D crystal structures to predict properties like dielectric constants and piezoelectric coefficients. Their multi-property pre-training (MPT) framework enabled accurate band gap predictions for 2D materials.